What is data analytics?

Data analytics is the process of analysing raw data in order to draw out meaningful, actionable insights.

It includes a range of tools, technologies, and processes used to find trends and solve problems by using data.

Data analytics can shape business processes, improve decision-making, and foster business growth.

Data analytics helps us to make sense of the past and to predict future trends and behaviours; rather than basing our decisions and strategies on guesswork, we're making informed choices based on what the data is telling us.

Furnished with the insights drawn from the data, businesses, and organizations are able to develop a much deeper understanding of their audience, their industry, and their company as a whole—and, as a result, are much better equipped to make decisions and plan ahead.

Data Science and Data Analytics are Different

Data science and data analytics are both incredibly important fields in the modern world. Both deal with the use of data to help make decisions and solve problems.

Data analytics and data science are two areas that are often confused with each other. The biggest difference between these two fields is their goals.

Data analytics focuses more on analysing an existing dataset, whereas data science focuses on creating new models to generate the best outcomes possible.

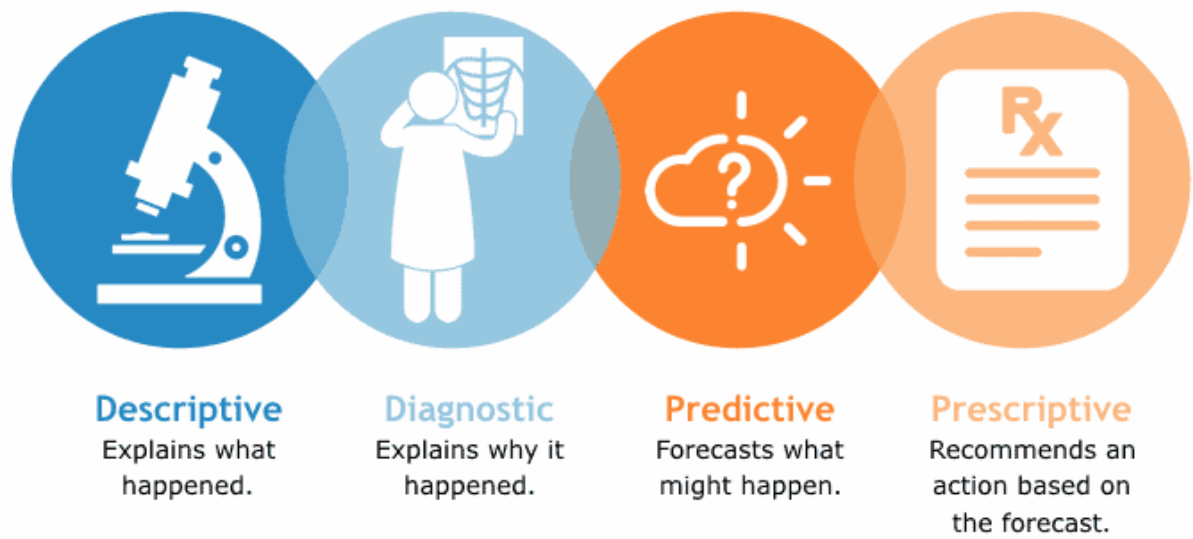
DIFFERENCES BETWEEN

DATA SCIENCE VS. DATA ANALYTICS

Feature	Data Science	Data Analytics
Coding Language	Python is the most commonly used language for data science along with the use of other languages such as C++, Java, Perl, etc.	The Knowledge of Python and R Language is essential for Data Analytics.
Programming Skills	In- depth knowledge of programming is required for data science.	Basic Programming skills is necessary for data analytics.
Use of Machine Learning	Data Science makes use of machine learning algorithms to get insights.	Data Analytics doesn't makes use of machine learning.
Other Skills	Data Science makes use of Data mining activities for getting meaningful insights.	Hadoop Based analysis is used for getting conclusions from raw data.
Scope	Care for your body parts	The Scope of data analysis is micro i.e., small.
Goals	Data science deals with explorations and new innovations.	Data Analysis makes use of existing resources.
Data Type	Data Science mostly deals with unstructured data.	Data Analytics deals with structured data.
Statistical Skills	The statistical skills are necessary in the field of Data Science.	The statistical skills are of minimal or no use in data analytics.

What are the different types of data analysis?

The four main types of data analysis are:
descriptive, diagnostic, predictive, and prescriptive.



Descriptive analytics

What happened?

Descriptive analytics alters raw data from multiple data sources to give valuable insights into the past. However, these findings simply signal that something is wrong or right, without explaining **why?**

For example, Data analysts are working with an e-commerce marketing team to review sales data to identify the sales trends and patterns, you will see an increase or decrease in sales from last year, specifically in what region and by what percentage.

Common examples of Descriptive analytics are company reports that provide historical reviews like:

- Data Queries
- Reports
- Descriptive Statistics
- Data dashboard

Diagnostic analytics

Why did something happen?

Like descriptive analytics, diagnostic analytics also focuses on the past. However, these types of analyses look for cause and effect to illustrate ***why*** something happened. The objective is to compare past occurrences to determine causes. Diagnostic analytics can guide by helping to:

- *Identify outliers*
- *Isolate patterns*
- *Uncover relationships*

Back to our marketing example, now that you are aware that there is a drop in sales, you can identify the reason why there a sudden drop in sales? This may need additional examination for that might need to look at additional data points like website traffic, marketing budgets, product inventory availability and make a correlation to uncover relationships. Using more complex analytics, analysts may employ probability theory, regression analysis, or time series to isolate cause and effect relationships.

Common techniques used for Diagnostic Analytics are:

- Data discovery
- Data mining
- Correlations

Predictive analytics

What is LIKELY to happen?

Predictive analytics tells ***what** is likely to happen.* It uses the findings of descriptive and diagnostic analytics to detect clusters and exceptions and to predict future trends, which makes it a valuable tool for forecasting. It brings many advantages like sophisticated analysis based on machine learning or deep learning and a proactive approach that predictions enable.

It is important to keep in mind, however, that no analytics will be able to tell you exactly what WILL happen in the future. Predictive analytics put in perspective what **MIGHT** happen, providing respective probabilities of likelihoods given the variables that are being looked at.

Now that we know the reason for the drop in sales. Predictive analytics helps to find what would be the expected sale in the next month, quarter, or year, etc. The goal is to determine a trend, correlation, causation, or probability for the next purchase.

Techniques that are used for predictive analytics are:

- *Linear Regression*
- *Time Series Analysis and Forecasting*
- *Data Mining*

Basic Cornerstones of Predictive Analytics

- *Predictive modeling*
- *Decision Analysis and optimization*
- *Transaction profiling*

Prescriptive analytics

What action should be taken?

The purpose of prescriptive analytics is to literally prescribe *what action to take* to eliminate a future problem or take full advantage of a promising trend. It uses advanced tools and technologies, like machine learning, business rules, and algorithms, which makes it sophisticated to implement and manage. Besides, this state-of-the-art type of data analytics requires not only historical internal data but also external information due to the nature of the algorithms it's based on.

As part of prescriptive analytics, if sales are falling then you can make timely decisions such as to cut prices, market more, or discontinue the product. If an item is selling off the shelves, you can be sure to stock inventory accurately across channels.